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Pothole Detection with Image Segmentation for Advanced Driver Assisted Systems

H.K.I.S. Lakmal
Dept. of Electrical and Electronic Engineering
National Engineering Research and Development Center
Ja-Ela, Sri Lanka
isurushanakalakmal@gmail.com

Maheshi B. Dissanayake
Dept. of Electrical and Electronic Engineering
University of Peradeniya
Peradeniya, Sri Lanka
maheshid@ee.pdn.ac.lk

***Abstract*—It is common to see water trapped on roads after rain. Potholes are the main reason behind trapped rainwater on roads. In general, it is advisable to avoid these water pools when driving. This research presents a vision-based Advanced Driver Assistance System that detects the water pools on asphalt roads. The presented system will assist drivers to take evasive action. Since most of the automobiles come with a camera system, we propose image-based simple machine learning architecture to detect water pools. The paper analyses the contribution of edge features, texture features and Gabor filtered features for the detection of the water surfaces. Initially, these features are ranked according to their importance. Next three Random Forest Classifier models are created and trained using a random number of features from the feature set according to their ranks. Training accuracy of all 3 models varies from 0.94-0.99 indicating a promising outlook for the given application.**

Keywords—Pothole Detection, Image Segmentation, Water Surface Detection, Random Forest Classifier, Feature Selection, Gabor Filters.

I. INTRODUCTION

Potholes is a common observation in countryside roads and rainwater is trapped on roads due to potholes, cracks and other uneven road surface conditions. Since it is difficult to determine the depth of the potholes when they are filled with water, unsuspecting vehicles hitting them may get their suspension system and other mechanical systems damaged. Also, it may affect the safety of the passengers when hitting a pothole without slowing down. Alternatively, hitting potholes when they are filled with water creates water splashes. Sand and mud will be stuck under the automobile body parts along with water splashes which is the key reason for corrosion. Besides, mud comes with water splashes can cause malfunctioning of the sensors in vehicles with sophisticated sensor systems. Also, water splashes can affect pedestrians, cyclists, other automobiles and the environment. Therefore, it is always advisable to avoid potholes when driving. When developing an Advanced Driver Assisted System (ADAS) which can navigate automobiles avoiding potholes, the first step is implementing a system which can detect potholes, which is discussed in this paper.

Studies with similar applications can be found in the literature. Srivastava, Goyal and Ram [1] discuss the feasibility of pothole detection using automotive radar. Eriksson et al. [2] present a method of detecting and reporting of surface conditions on roads using accelerometer sensors fixed to vehicles where potholes and severe road conditions are estimated using the inherent mobility of the participating vehicles. A simple machine learning approach is used to estimate the potholes using the accelerometer data. Similarly, Strazdins, Mednis, Kanonirs, Zviedris and Selavo [3] explain the possibility of implementing a road condition monitoring system using Android smartphones and [4] discusses the implementation of such a system. A study by Yu and Salari [5] discusses a pavement distress inspection system using a laser imaging technique. After capturing the laser images, the pothole severity and crack type are estimated by providing input to a feed-forward neural network for pothole severity and crack type classification. Chang, Chang and Liu [6] also explain an approach of 3D laser scanning similar to [5]. According to the study by Jo and Ryu [7] automatic detection systems, such as vibrations based or laser scanning based systems, are insufficient to detect potholes correctly and inexpensively and hence propose a road condition monitoring system using a camera. Koch and Brilakis [8] also explains a similar system which detects the potholes in asphalt pavement images. Emir, Samir and Huseinovic [9] also explain an image process approach to pothole detection using spectral clustering. Since all image processing applications discussed above are focused on road condition monitoring, it was considered the roads are in dry road conditions only. The effect of wet roads where potholes are filled with water was not considered in the studies presented above. A study by Quaio, Wu, He, Li and Peng [10] discusses a deep learning approach for splashed water detection on roads yet this application is different from the pothole detection when there is water.

Distinctive from existing research, this paper presents an application for pothole detection which can be used to detect potholes filled with water. The method presented, extracts features from input images, rank the features according to their importance and trains a segmentation machine using random forest classifier. The trained model is able to successfully recognize potholes with nearly 0.94-0.99 accuracy.

II. BACKGROUND

Water collected pothole detection is an image segmentation problem in machine vision. Image segmentation can be divided into two classes as instance segmentation and semantic segmentation where the latter is used in the work presented in this manuscript. Although there are several ways to implement semantic segmentation, using Machine Learning (ML) algorithm is simple and versatile for this purpose. The performance success of the designed ML model will heavily depend on feature extraction and feature selection form the input images.

One of the easiest approaches to detect discontinuities in an image is edge filtering. Roberts Edge filter detects simple edges in an image whereas any Edge filter uses a multi-stage algorithm to detect a wider range of edges. Sobel filter is

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another edge detection algorithm which emphasizes edges in an image. Scharr filter is a filtering method that is used to identify and highlight gradient edges using the first derivative along the x-axis and y-axis independently in the image. Performance this filter is quite similar to the Sobel filter. Prewitt filter is also a popular edge detection algorithm. It computes an approximation of the gradient of the image intensity. In general, all these edge filters estimate the edges in the image, using different approaches. Hence the features extracted from each edge filter is different from the other and they can be used to detect the boundary of the water-spots.

Beside edge features, we can obtain other texture and pattern feature through filters such as Median, Variance, and Gaussian filters. The Median filter produces a noise-reduced image output while Variance filter provides the information about the amount of variation in the image. Further, the variation filter can be used as a dimensionality reduction algorithm as well. A Gaussian filter is a linear filter and it is used to blur the image and enhance image structure at different scales. Small or large blur can be applied to the image based on the standard deviation used.

Gabor filter is a popular linear filter used for texture analysis. It used to examine the specific frequency content in the image in specific directions. Parameters such as the wavelength of the sinusoidal component (λ), the orientation of the normal to the parallel stripes (θ), the standard deviation of the Gaussian envelope (σ), and the spatial aspect ratio (γ) control the shape and size of the Gabor filter. Due to the inherent characteristics of the Gabor filters, it is used as a popular feature extraction method in computer vision applications [11].

After extracting feature, feature selection should be carried out which remove the features that are either redundant or irrelevant without incurring much loss of information. This can be achieved easily after ranking the features based on the feature importance.

Many machine learning algorithms such as logistic regression, support vector machine, Naive Bayes classifier, and decision trees which can be used for classification. However, the Random Forest Classifier (RFC)algorithm is widely used for classification problems with smaller datasets as it suffers less from overfitting. RFC consists of a group of relatively uncorrelated decision trees which operate as an ensemble. In RFC, each decision tree generates an independent class prediction and the class with the most votes becomes the output of the RFC. Since uncorrelated models can produce ensemble predictions that are more accurate than any of the individual predictions. Decision trees in the RFC allow splitting the data or observation in a way that, the resulting groups are as different from each other as possible and the members of each resulting subgroup are as similar to each other as possible at each node based on the features. Therefore, the importance of each feature is derived based on the purity of each subgroup. This property of RFC can itself be used for the feature selection process as well.

III. Methodology

The methodology of this research can be divided into four stages as preparation of the dataset, feature extraction from images, feature selection and building the model. All coding and implementation work in the above stages were carried out on an ASUS X556UQK computer with Intel(R) Core (TM) i5-7200U CPU, 8GB RAM and Windows 10 Pro 64-bit (10.0, Build 19041) operating system.

A. Dataset Preparation

The first step of this application was generating the dataset. Therefore, a set of images with potholes were collected and those were resized to the width of 400 pixels and the height of 300 pixels. Since RFC is a supervised machine learning algorithm, it was necessary to prepare both input and label datasets for training. In this case, a pixel-wise masked image should be generated as the label for each input image. Since 400 x 300 pixels images with potholes were used as input images, similar-sized images were created with a white mask on the potholes and black mask on the remaining area. Fig. 1 (a) shows such an input image and Fig. 1 (b) shows the corresponding label image (i.e. masked image). Masked images were created manually using the INKSCAPE 0.92.4 open-source vector graphics editor. Eventually, the dataset consisted of 40 number of input images and corresponding 40 number of label images.

B. Feature Extraction

Since extraction was a prolonged process as each feature should be extracted from every image and stored them discretely. Before starting the feature extraction process, all input images were converted to greyscale to reduce the computational cost later in the ML model training. Yet, it should be stated that if RGB features were employed the performance may have improved further in the proposed system. After the conversion is completed it was saved in a Python-Pandas data frame as a feature since the greyscale image itself was used as a feature for model tuning.

Next, by using appropriate filters other feature sets were extracted from the grayscale image. At this stage, the grayscale image was favoured against RGB image, as processing grayscale was computationally less expensive compared to the other. Canny, Roberts, Sobel, Scharr and Prewitt edge filter was applied to the greyscale image to obtained the edge-based feature sets. Another set of features were extracted using noise reduction and blurring process, utilizing Gaussian, Median and Variance filters. All these extracted features also saved under different column names in the same Pandas data frame.

Final features extracted were the texture features of the image. Gabor kernel (9x9) was used for texture feature extraction, while the shape and the size of the Gabor function were adjusted to get the best feature set. Table 1 shows the Gabor parameters used and by varying these parameters 32 different Gabor filters were created for texture feature extraction. The output feature sets of the Gabor filters were also saved in the Pandas data frame under 32 number of separate columns.



(a) (b)

Fig. 1. (a) 400 x 300 pixel input image, (b) Corresponding label image.

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TABLE I. GABOR FUNCTION PARAMETERS

Symbol	Parameter	Values
λ	Wavelength of the sinusoidal component	0, π/4, π/2, π
θ	Orientation of the normal to the parallel stripes	0, π/4
σ	Standard deviation of the Gaussian envelope	1, 3
γ	Spatial aspect ratio	0.05, 0.5

These extracted 42 number of different features were used to train the ML model. Since all input images were converted to numerical features, the binary-label images were also converted to the numerical format.

C. *Feature Selection*

As the next step features extracted were ranked based on their importance. As explained earlier, the feature importance could be calculated during the training of the RFC. Therefore, an initial RFC was trained using only 20% of the dataset to evaluate the feature importance for the model training. Feature importance ranking obtained was plotted in the ascending order in Fig. 2. It shows only the features whose importance is greater than the mean importance of all the features since the importance of remaining features were significantly smaller (i.e. out of 42 the number of features, only 19 features had contributed for detecting water surfaces in an image). Even though all 19 features (or even all 42 features) could be used to build and train one RFC, the training process would be highly time and resource consuming. Therefore, it was decided to build several RFC models by selecting randomly a different number of features and evaluate the accuracy against the computation time to select the best-suited model. 9, 6 and 3 features with the highest feature importance were selected to build RFC model 1, model 2 and model 3 respectively as shown in Table 2.

D. *Building and Training the RFC model*

Since three RFC models were selected, the next step was to build RFC models and training them on the original dataset. The original dataset was divided into two datasets as training and testing datasets where 60% of the original dataset represents the training dataset and rest represent the testing dataset. Total computation time which is taken to build and train models were used as an indicator for the computational complexity of the model. Also, training and testing accuracies of the models were calculated to evaluate the RFC models. Accuracy and computation time of each model is tabulated in Table 3.

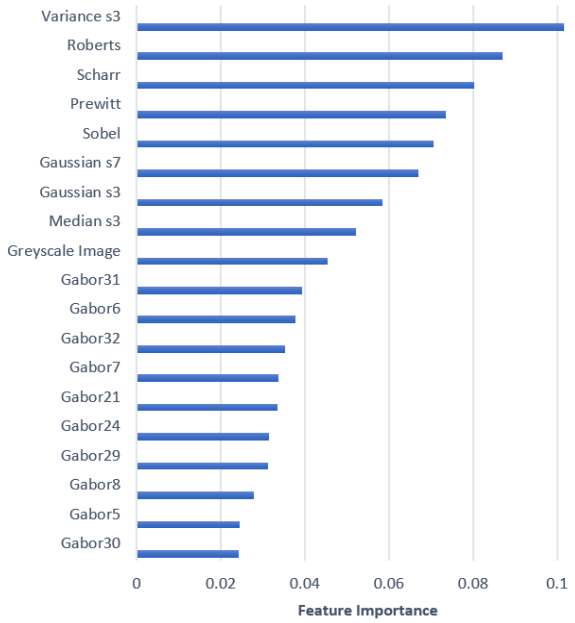


Fig. 2. Calculated importance of features.

TABLE II. FEATURE SELECTION FOR RFC MODELS

RFC model	Selected Features
Model 1	Variance s3, Roberts, Scharr, Prewitt, Sobel, Gaussian s7, Gaussian s3, Median s3, Greyscale Image
Model 2	Variance s3, Roberts, Scharr, Prewitt, Sobel, Gaussian s7
Model 3	Variance s3, Roberts, Scharr

According to Table 3, it is evident that both the computational time as well as the model accuracy decrease when the number of features used for the model training and testing decreases. Fig. 3. compares the training and testing accuracy of the models.

IV. RESULTS AND DISCUSSION

According to the performance of all three RFC models, model accuracy can be increased by increasing the number of features used to train the RFC model subjected to an increase in the computational complexity. Table 4. shows the prediction of three sample images and their corresponding prediction accuracy. Results presented in Table 4 further clarify the previous argument, that is the accuracy of the RFC model increases with the increase in the number of features taken to build the model since Model 2 gives a more accurate prediction compared to Model 3 and Model 1 gives a more improved prediction than Model 2.

Yet, having more features lead to higher computational cost, especially when working with larger data sets. Therefore, it is always advisable to reduce the computational load as much as possible when and where possible.

TABLE III. ACCURACIES OF TRAINED RFC MODELS

RFC model	Accuracy		Training time / min	Testing time / sec
	Training	*Testing*		
Model 1	0.9984	0.8775	49.6	8.5938
Model 2	0.9879	0.8579	31.4	6.4844
Model 3	0.9438	0.8311	21.8	5.3438

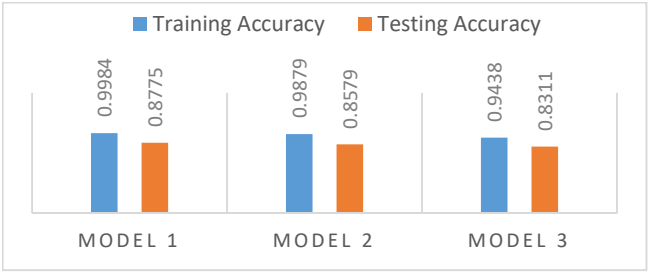
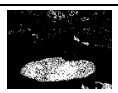
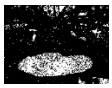


Fig. 3. Training and testing accuracy of models.

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TABLE IV PREDICTION OF RFC MODELS FOR SAMPLE IMAGES

		Sample 1	Sample 2	Sample 3
Original				
Target				
Model 1	***Prediction***			
	Accuracy	0.8397	0.7841	0.8493
Model 2	***Prediction***			
	Accuracy	0.8197	0.7543	0.8293
Model 3	***Prediction***			
	Accuracy	0.7362	0.6917	0.6948

Further, all three RFC models presented have shown decent accuracy. Even the Model 3 with only three features (Variance s3, Roberts, Scharr) functions with a considerable level of accuracy. Hence, for a better performing ML model, it is essential to optimize the trade-off between the number of features and the required accuracy as well as computation time.

V. CONCLUSION

The first step of building a successful machine learning model for image segmentation is extracting correct features from the input image. The presented research focuses on water-filled pothole detection using computer vision as a driver assistance mechanism. This paper presents several different features that can be utilized to detect the water surfaces and to segment water area in an input image. Furthermore, we have used Random Forest Classifier for feature ranking and model training for the water surface segmentation. With the proposed architecture we were able to reach training accuracy of 0.998 and testing accuracy of 0.877.

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